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Opgave: Vierkantswortel ($\mathbb{C}$) van $20i + 21$ nemen.
 $20i + 21 = 21 + 20i$

$$\Rightarrow 21 + 20i = (x + yi)^2 = x^2 - y^2 + 2xyi$$

$$\Rightarrow \begin{cases} x^2 - y^2 = 21 \\ 2xy = 20 \end{cases}$$

$$\Leftrightarrow \begin{cases} x^2 - y^2 = 21 \\ x^2(-y)^2 = -100 \end{cases}$$

$$\Rightarrow \lambda^2 - 21\lambda - 100 = 0$$

$$\Rightarrow D = 441 - 4 \cdot 1 \cdot (-100) = 841 \Rightarrow \sqrt{D} = 29$$

$$\Rightarrow \begin{cases} \lambda_1 = \frac{21 + 29}{2} = 25 \\ \lambda_2 = \frac{21 - 29}{2} = -4 \end{cases}$$

$$\Rightarrow \begin{cases} x^2 = 25 \stackrel{b>0}{\Leftrightarrow} x = 5 \\ y^2 = 4 \Leftrightarrow y = 2 \end{cases}$$

$$\Rightarrow \begin{cases} w_1 = 5 + 2i \\ w_2 = -5 - 2i \end{cases}$$

References